**Data Cleaning**

## 1. Missing Vendor Integrity Check

**TITLE**

**Vendors with No Matching Audit Record**

**WHAT I DID**

Identified vendor audit records that do not have a matching vendor in the main vendor table using a LEFT JOIN

```sql
-- 1. Audits with no matching Vendor
SELECT a.vendor_id
FROM vendor_audit a
LEFT JOIN tprm_vendors v ON a.vendor_id = v.vendor_id
WHERE v.vendor_id IS NULL;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

| vendor_id |
|---|

**INSIGHT**

This reveals orphan audit records that do not map to any existing vendor, indicating potential data integrity issues that could affect reporting accuracy.

## 2. Invalid Score Validation

**TITLE:**

**Detection of Invalid Risk Scores**

**WHAT I DID**

Checked for unrealistic cyber and financial risk scores outside the valid range of 0 to 100.

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```sql
-- 2. Check for scores outside the 0 to 100 range
SELECT * FROM vendor_audit
WHERE cyber_score > 100 OR cyber_score < 0
   OR Financial_score > 100 OR Financial_score < 0;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

| audit_id | vendor_id | audit_date | financial_score | cyber_score | compliance_score | Overall_risk_complaince |
|---|---|---|---|---|---|---|
| NULL | NULL | NULL | NULL | NULL | NULL | NULL |

**INSIGHT**

Several records contain invalid score values, which may distort risk analysis and require correction before any risk modeling or reporting.

## 3. Missing Data Detection

## TITLE

**Identifying Missing Audit Data**

**WHAT I DID**

Queried the audit table to detect missing values in critical fields such as audit date and compliance scores.

```sql
-- 3.Checking of Rows with missing data in vendor audits table
SELECT * FROM vendor_audit
WHERE Overall_risk_complaince IS NULL
   OR audit_date IS NULL;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

| audit_id | vendor_id | audit_date | financial_score | cyber_score | compliance_score | Overall_risk_complaince |
|---|---|---|---|---|---|---|
| NULL | NULL | NULL | NULL | NULL | NULL | NULL |

## INSIGHT

Missing audit information reduces data reliability and can lead to incomplete risk assessments if not addressed.

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## 4. Duplicate Records Detection

### TITLE

**Identifying Duplicate Audit Records**

**WHAT I DID**

Used GROUP BY and COUNT logic to detect duplicate vendor audit entries based on vendor ID and audit date.

```sql
-- 6. checking duplication in vendor audit
SELECT vendor_id, audit_date, COUNT(*)
FROM vendor_audit
GROUP BY vendor_id, audit_date
HAVING COUNT(*) > 1;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

| vendor_id | audit_date | COUNT(*) |
|---|---|---|
| 500 | 1/2/2023 | 2 |
| 500 | 1/7/2023 | 2 |
| 500 | 6/26/2025 | 4 |
| 500 | 3/21/2024 | 4 |
| 500 | 8/4/2023 | 2 |
| 500 | 9/29/2023 | 3 |
| 500 | 1/28/2023 | 2 |
| 500 | 9/3/2023 | 2 |

## INSIGHT

Duplicate audit records were found, which may inflate risk exposure metrics and lead to inaccurate reporting if not removed.

## 5. Future Date Logic Check

### TITLE

**Detection of Invalid Audit Dates**

**WHAT I DID**

Identified records where audit dates are set in the future, which violates logical data constraints.

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```sql
-- 8. Date logic auditing
SELECT * FROM vendor_audit
WHERE audit_date > CURRENT_DATE;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

| audit_id | vendor_id | audit_date | financial_score | cyber_score | compliance_score | Overall_risk_complaince |
|---|---|---|---|---|---|---|
| NULL | NULL | NULL | NULL | NULL | NULL | NULL |

## INSIGHT

Future dated audit entries indicate data entry errors or system issues, which must be corrected to maintain timeline accuracy in reporting.